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Question Answer System Implementation

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In [20]: !pip install sentence-transformers faiss-cpu

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: sentence-transformers in ./local/lib/python3.13/site-packages (6.0.1)
Requirement already satisfied: faiss-cpu in ./local/lib/python3.13/site-packages (1.15.0)
Requirement already satisfied: transformers<6.0.0,>=5.0.0 in ./local/lib/python3.13/site-packages (from sentence-transformers) (5.16.1)
Requirement already satisfied: tokenizers>=0.19 in ./local/lib/python3.13/site-packages (from sentence-transformers) (0.23.1)
Requirement already satisfied: huggingface-hub<2.0.0,>=1.3.0 in ./local/lib/python3.13/site-packages (from sentence-transformers) (1.28.0)
Requirement already satisfied: torch>=2.2 in ./local/lib/python3.13/site-packages (from sentence-transformers) (2.13.0)
Requirement already satisfied: numpy>=1.24.0 in ./local/lib/python3.13/site-packages (from sentence-transformers) (2.5.1)
Requirement already satisfied: scikit-learn>=1.1.0 in ./local/lib/python3.13/site-packages (from sentence-transformers) (1.9.0)
Requirement already satisfied: scipy>=1.0.0 in ./local/lib/python3.13/site-packages (from sentence-transformers) (1.18.0)
Requirement already satisfied: typing_extensions>=4.10.0 in /opt/anaconda/lib/python3.13/site-packages (from sentence-transformers) (4.12.2)
Requirement already satisfied: tqdm>=4.0.0 in /opt/anaconda/lib/python3.13/site-packages (from sentence-transformers) (4.67.1)
Requirement already satisfied: click<9.0.0,>=8.4.2 in ./local/lib/python3.13/site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (8.4.2)
Requirement already satisfied: filelock>=3.10.0 in /opt/anaconda/lib/python3.13/site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (3.17.0)
Requirement already satisfied: fsspec>=2023.5.0 in /opt/anaconda/lib/python3.13/site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (2025.3.2)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.2 in ./local/lib/python3.13/site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (1.6.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /opt/anaconda/lib/python3.13/site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (0.28.1)
Requirement already satisfied: packaging>=20.9 in /opt/anaconda/lib/python3.13/site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (24.2)
Requirement already satisfied: pyyaml>=5.1 in /opt/anaconda/lib/python3.13/site-packages (from huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (6.0.2)
Requirement already satisfied: anyio in /opt/anaconda/lib/python3.13/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (4.7.0)
Requirement already satisfied: certifi in /opt/anaconda/lib/python3.13/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (2025.4.26)
Requirement already satisfied: httpcore==1.* in /opt/anaconda/lib/python3.13/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (1.0.9)
Requirement already satisfied: idna in /opt/anaconda/lib/python3.13/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (3.7)
Requirement already satisfied: h11>=0.16 in /opt/anaconda/lib/python3.13/site-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (0.16.0)

Requirement already satisfied: regex>=2025.10.22 in ./local/lib/python3.13/site-packages (from transformers<6.0.0,>=5.0.0->sentence-transformers) (2026.8.31)

Requirement already satisfied: typer in /opt/anaconda/lib/python3.13/site-packages (from transformers<6.0.0,>=5.0.0->sentence-transformers) (0.9.0)

Requirement already satisfied: safetensors>=0.8.0 in ./local/lib/python3.13/site-packages (from transformers<6.0.0,>=5.0.0->sentence-transformers) (0.8.0)

Requirement already satisfied: joblib>=1.4.0 in ./local/lib/python3.13/site-packages (from scikit-learn>=1.1.0->sentence-transformers) (1.5.3)

Requirement already satisfied: narwhals>=2.0.1 in ./local/lib/python3.13/site-packages (from scikit-learn>=1.1.0->sentence-transformers) (2.24.0)

Requirement already satisfied: threadpoolctl>=3.5.0 in /opt/anaconda/lib/python3.13/site-packages (from scikit-learn>=1.1.0->sentence-transformers) (3.5.0)

Requirement already satisfied: setuptools>=77.0.3 in ./local/lib/python3.13/site-packages (from torch>=2.2->sentence-transformers) (83.0.0)

Requirement already satisfied: sympy>=1.13.3 in /opt/anaconda/lib/python3.13/site-packages (from torch>=2.2->sentence-transformers) (1.13.3)

Requirement already satisfied: networkx>=2.5.1 in /opt/anaconda/lib/python3.13/site-packages (from torch>=2.2->sentence-transformers) (3.4.2)

Requirement already satisfied: jinja2 in /opt/anaconda/lib/python3.13/site-packages (from torch>=2.2->sentence-transformers) (3.1.6)

Requirement already satisfied: cuda-toolkit==13.0.3 in ./local/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch>=2.2->sentence-transformers) (13.0.3.0)

Requirement already satisfied: cuda-bindings<14,>=13.0.3 in ./local/lib/python3.13/site-packages (from torch>=2.2->sentence-transformers) (13.3.1)

Requirement already satisfied: nvidia-cudnn-cu13==9.20.0.48 in ./local/lib/python3.13/site-packages (from torch>=2.2->sentence-transformers) (9.20.0.48)

Requirement already satisfied: nvidia-cusparselt-cu13==0.8.1 in ./local/lib/python3.13/site-packages (from torch>=2.2->sentence-transformers) (0.8.1)

Requirement already satisfied: nvidia-nccl-cu13==2.29.7 in ./local/lib/python3.13/site-packages (from torch>=2.2->sentence-transformers) (2.29.7)

Requirement already satisfied: nvidia-nvshmem-cu13==3.4.5 in ./local/lib/python3.13/site-packages (from torch>=2.2->sentence-transformers) (3.4.5)

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Requirement already satisfied: nvidia-cublas==13.1.1.3.* in ./local/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch>=2.2->sentence-transformers) (13.1.1.3)

Requirement already satisfied: nvidia-cuda-nvrtc==13.0.88.* in ./local/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch>=2.2->sentence-transformers) (13.0.88)

Requirement already satisfied: nvidia-cuda-runtime==13.0.96.* in ./local/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch>=2.2->sentence-transformers) (13.0.96)

Requirement already satisfied: nvidia-cufft==12.0.0.61.* in ./local/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch>=2.2->sentence-transformers) (12.0.0.61)

Requirement already satisfied: nvidia-nvjitlink<14,>=13.0.88 in ./local/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_s

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system == "Linux"->torch>=2.2->sentence-transformers) (13.3.33)
Requirement already satisfied: nvidia-cublas==1.15.1.6.* in ./local/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch>=2.2->sentence-transformers) (1.15.1.6)
Requirement already satisfied: nvidia-cuda-cupti==13.0.85.* in ./local/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch>=2.2->sentence-transformers) (13.0.85)
Requirement already satisfied: nvidia-curand==10.4.0.35.* in ./local/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch>=2.2->sentence-transformers) (10.4.0.35)
Requirement already satisfied: nvidia-cusolver==12.0.4.66.* in ./local/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch>=2.2->sentence-transformers) (12.0.4.66)
Requirement already satisfied: nvidia-cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch>=2.2->sentence-transformers) (12.6.3.3)
Requirement already satisfied: nvidia-nvtx==13.0.85.* in ./local/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch>=2.2->sentence-transformers) (13.0.85)
Requirement already satisfied: cuda-pathfinder>=1.4.2 in ./local/lib/python3.13/site-packages (from cuda-bindings<14,>=13.0.3->torch>=2.2->sentence-transformers) (1.6.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/anaconda/lib/python3.13/site-packages (from sympy>=1.13.3->torch>=2.2->sentence-transformers) (1.3.0)
Requirement already satisfied: sniffio>=1.1 in /opt/anaconda/lib/python3.13/site-packages (from anyio->httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=1.3.0->sentence-transformers) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /opt/anaconda/lib/python3.13/site-packages (from jinja2->torch>=2.2->sentence-transformers) (3.0.2)

```

In [6]: `!pip install scikit-learn`

```

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: scikit-learn in ./local/lib/python3.13/site-packages (1.9.0)
Requirement already satisfied: numpy>=1.24.1 in ./local/lib/python3.13/site-packages (from scikit-learn) (2.5.1)
Requirement already satisfied: scipy>=1.10.0 in ./local/lib/python3.13/site-packages (from scikit-learn) (1.18.0)
Requirement already satisfied: joblib>=1.4.0 in ./local/lib/python3.13/site-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: narwhals>=2.0.1 in ./local/lib/python3.13/site-packages (from scikit-learn) (2.24.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in /opt/anaconda/lib/python3.13/site-packages (from scikit-learn) (3.5.0)

```

In [2]: `!pip install olefile`

```

Defaulting to user installation because normal site-packages is not writeable
Collecting olefile
  Downloading olefile-0.47-py2.py3-none-any.whl.metadata (9.7 kB)
Downloading olefile-0.47-py2.py3-none-any.whl (114 kB)
Installing collected packages: olefile
Successfully installed olefile-0.47

```

```

In [15]: import os
import olefile

class Extract:
    def extract_text_pure_python_doc(self, file_path):
        if not olefile.isOleFile(file_path):
            raise ValueError("This file is not a valid legacy OLE .doc file")

        ole = olefile.OleFileIO(file_path)

        # Legacy .doc text is stored inside the 'WordDocument' stream
        if ole.exists('WordDocument'):
            doc_stream = ole.openstream('WordDocument')
            raw_bytes = doc_stream.read()

            # Decode text by stripping out control characters and non-ascii
            # This extracts raw strings directly from the binary data blocks
            text_pieces = []
            current_piece = []

            for b in raw_bytes:
                # Keep standard printable text characters and line breaks
                if 32 <= b < 127 or b in (10, 13, 9):
                    current_piece.append(chr(b))
                else:
                    if current_piece:
                        clean_str = "".join(current_piece).strip()
                        # Only keep blocks that actually contain words
                        if len(clean_str) > 3:
                            text_pieces.append(clean_str)
                        current_piece = []

            if current_piece:
                text_pieces.append("".join(current_piece))

            ole.close()
            return "\n".join(text_pieces)
        else:
            ole.close()
            raise KeyError("Could not find a valid WordDocument text stream")

e1=Extract()
# Execute with your path
document_path = "/home/matlab/Downloads/sample.doc"
document_text = e1.extract_text_pure_python_doc(document_path)
print(f"Extracted {len(document_text)} characters from the legacy document")

```

Extracted 32331 characters from the legacy document.

```

In [4]: def chunk_text(text, chunk_size=500, overlap=50):
        chunks = []
        start = 0

```

```

while start < len(text):
    end = start + chunk_size
    chunks.append(text[start:end])
    start += chunk_size - overlap # Move forward while keeping the overlap
return chunks

# Split the text into chunks
chunks = chunk_text(document_text, chunk_size=500, overlap=50)
print(f"Created {len(chunks)} text chunks.")

```

Created 72 text chunks.

```

In [7]: import numpy as np
import faiss
from sklearn.feature_extraction.text import TfidfVectorizer

# Initialize a local vectorizer (Requires 0 internet access)
# max_features sets the vector dimension size (e.g., 512 dimensions)
vectorizer = TfidfVectorizer(max_features=512, stop_words='english')

# Learn vocabulary and convert text chunks to a sparse matrix
tfidf_matrix = vectorizer.fit_transform(chunks)

# Convert the sparse matrix to a dense float32 array for FAISS compatibility
chunk_embeddings = tfidf_matrix.toarray().astype('float32')

# Get the fixed vector dimension size
dimension = chunk_embeddings.shape[1]

# Initialize a standard flat L2 distance FAISS index
index = faiss.IndexFlatL2(dimension)

# Add the local chunk vectors to the index
index.add(chunk_embeddings)

print(f"Successfully vectorized and stored {index.ntotal} vectors in FAISS")

```

Successfully vectorized and stored 72 vectors in FAISS completely offline.

```

In [18]: class Retrieval:
def retrieve_relevant_chunks(self, question, k=3):
    # Vectorize the user's question using the local trained vectorizer
    question_matrix = vectorizer.transform([question])
    question_embedding = question_matrix.toarray().astype('float32')

    # Perform similarity search in FAISS
    distances, indices = index.search(question_embedding, k)

    # Extract the actual text chunks matching the top indices
    retrieved_contexts = [chunks[idx] for idx in indices[0] if idx != -1]
    return retrieved_contexts
print("Local offline retrieval system is ready.")

```

```

In [22]: r1=Retrieval()

# Select one of the appropriate test questions
user_question = "What is the cost for color printing of figures?"

# Retrieve matching chunks from your local TF-IDF FAISS index
matched_chunks = r1.retrieve_relevant_chunks(user_question, k=1)

```

```
print(f"Testing Question: {user_question}\n")
print("Top Extracted Chunk from FAISS:")
print("-" * 50)
for i, chunk in enumerate(matched_chunks, 1):
    print(chunk)
```

Testing Question: What is the cost for color printing of figures?

Top Extracted Chunk from FAISS:

. Do not abbreviate
Table.

Color printing of figures is available, but is billed to the authors (approximately \$1300, depending on the number of figures and number of pages containing color). Include a note with your final paper indicating that you request color printing. Do not use color unless it is necessary for the proper interpretation of your figures. If you want reprints of your color article, the reprint order should be submitted promptly. There